DIBYA DEEPTA MISHRA

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PhD in Economics and MS in Computer Science. Apply econometrics and machine learning to pricing, experimentation, and product decisions.

EXPERIENCE

Cornerstone Research, Associate — Chicago, IL · Jan 2025 to Sep 2026

- Estimated demand elasticities using structural demand models, calculated margins from accounting statements, and used regressions to estimate "but-for" price effects for market definition and competition assessments; supported merger reviews with a combined multibillion-dollar transaction value that regulators cleared after extended review.
- Applied difference-in-differences, synthetic controls, and high-dimensional fixed-effects regression to datasets with 10M+ observations to assess price effects in healthcare and technology litigation; findings were incorporated into expert reports.
- Modified a theoretical pricing model to assess "but-for" prices and damages claims; the model was incorporated into an expert report supporting testimony in antitrust litigation.
- Managed 2 to 3 analysts across cases and mentored them on econometric methods, reproducible workflows, code review, and version control to support expert-report preparation; identified and corrected code implementation issues in analyses used in expert reports.
- Tracked healthcare cases, evaluated the conduct at issue and potential damages, and briefed partners monthly to prioritize outreach; contributed economic analyses to healthcare and technology pitches that helped secure client engagements.

Amazon Retail Pricing, Economist Intern — Seattle, WA · May 2023 to Aug 2023

- Combined product-text embeddings with gradient boosting to forecast product-level demand and inform discount amounts and bundle prices; reduced forecast RMSE by 5% relative to standard logit demand models.
- Used demand models to evaluate discounts and bundle prices with the objective of maximizing long-term customer revenue; presented recommendations internally to inform future A/B tests.

Indian School of Business, Research Associate — Hyderabad, India · Sep 2017 to Apr 2019

- Estimated consumer-choice and firm-entry models using healthcare claims data from India to assess government reimbursement policy; contributed the analysis to a working paper.
- Coauthored Incentives in Healthcare to examine incentive design for doctors, patients, nurses, other healthcare workers, and regulators; the case was published through Harvard Business Publishing and used in MBA teaching.

RESEARCH

SPAR (Supervised Program for Alignment Research), Research Fellow; selected from about 6,000 applicants · Sep 2026 to Present

- Estimate discrete-choice demand models to measure customers' willingness to pay for AI capability and safety and inform model and product design; the project has produced preliminary estimates and a working draft intended for circulation.

SKILLS

Econometrics and experimentation: Causal inference, A/B test analysis, difference-in-differences, demand estimation, instrumental variables, matching, synthetic controls, high-dimensional fixed-effects regression, discrete choice, price elasticity, counterfactual simulation, double machine learning, doubly robust estimation.

Machine learning: Feature engineering, model validation, gradient boosting, neural networks, product-text embeddings, analysis of language-model representations.

Programming and data: Python (pandas, NumPy, scikit-learn, PyTorch), SQL, R, Git, Stata, Julia.

Cloud and distributed tools: AWS EC2 and SageMaker; Apache Spark.

EDUCATION

Rice University, PhD in Economics; Industrial Organization and Structural Econometrics 2024

Relevant coursework: Advanced Econometrics; Computational Economics; Labor Economics; Industrial Organization I and II; Corporate Finance; Asset Pricing; Financial Econometrics.

Georgia Institute of Technology, MS in Computer Science; Machine Learning and AI 2024

Relevant coursework: Machine Learning; Machine Learning for Trading; AI Ethics; Natural Language Processing; Algorithms; Stochastic Optimization; Data and Visual Analytics; High-Dimensional Data Analytics.

London School of Economics, MSc in Econometrics and Mathematical Economics 2017

Indian Institute of Management Calcutta, PGDM 2014

Indian Institute of Technology Jodhpur, BTech in Electrical Engineering 2012

PAPERS

Published

AI-assisted teams outperform AI-led teams but not human-only teams in assessing research reproducibility in quantitative social science | With Abel Brodeur et al. | PNAS, 2026

The Wizard of OZ (Opportunity Zones): Spatial Spillovers in Place-Based Programs | With Robin C. Sickles and Yanfei Sun | Seven Decades of Econometrics and Beyond, Springer, 2025

Working

Quality Considerations and Universal Healthcare | Sole author | Working paper, 2026

Spectrum Allocation to Public Operators: A Structural Analysis of Set-Asides versus Subsidies in Indian Telecom Auctions | With Rajarshi Bhowal and Nurdaulet Menglibayev | Under review, 2026

Pennies for Pads | With Ritika Sethi | Under review, 2026

Do Language Models Represent Utility-Maximizing Choice? | Sole author | Under review, 2026